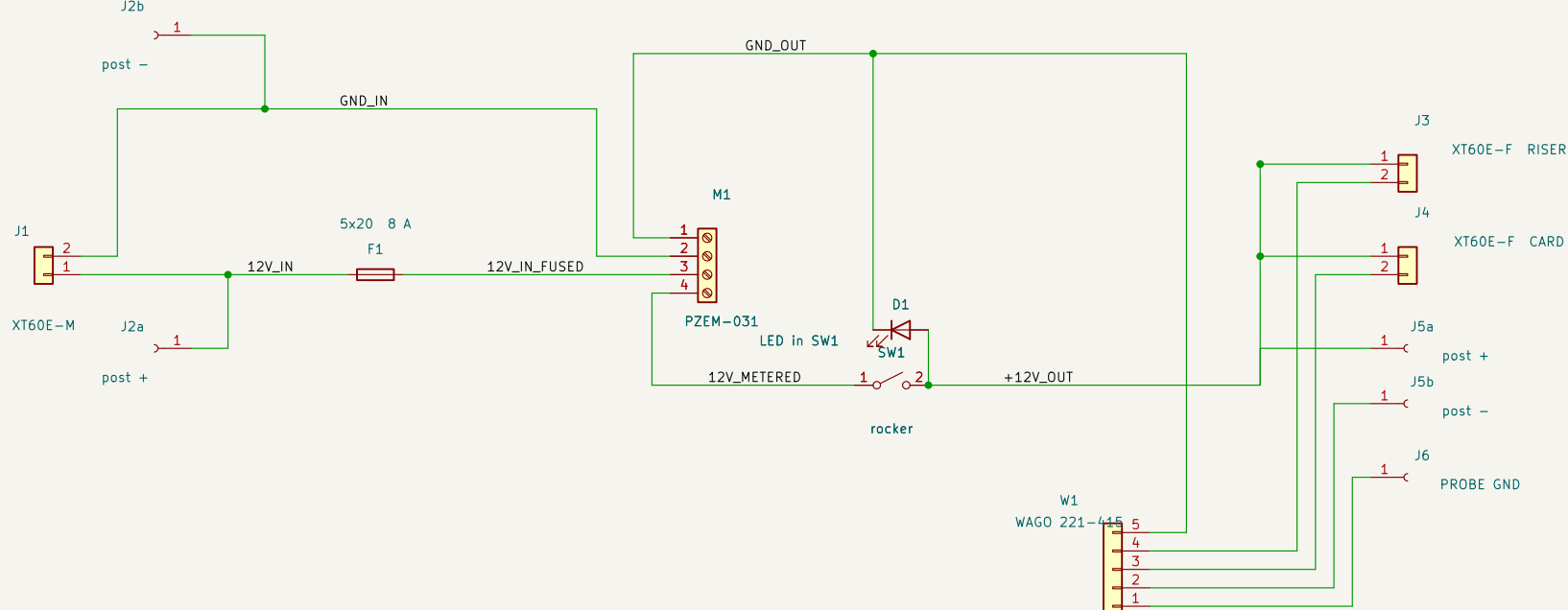


PCIe GPU test rig – power box wiring

Rev C 2026-09-28. 12V IN -> F1 -> M1 -> SW1 -> outputs. One common ground bus on the meter's LOAD- (W1).



Ref	Part
J1	XT60E-M panel mount, male – 12 V IN. PSU lead ends in a female XT60H
J2a/b	4 mm binding posts, red / black – 12 V IN, parallel with J1
F1	5x20 mm panel fuse holder, 8 A fast-blow glass fuse
M1	Peacefair PZEM-031 meter, 8-100 V / 0-20 A, built-in shunt
SW1	Amppper 20 mm illuminated rocker, 12 V 20 A, 3-pin
D1	the dot LED inside SW1 (yellow pin = LED -)
W1	WAGO 221-415 five-port lever nut – ground bus
J3	XT60E-F panel mount, female – OUT, RISER
J4	XT60E-F panel mount, female – OUT, CARD (optional 2nd, parallel)
J5a/b	4 mm binding posts, red / black – OUT +/-
J6	4 mm binding post, black – PROBE GND (scope / DMM reference)

Where
rear wall
rear wall
rear wall
top deck
top deck
-
inside, on the base floor
right wall
right wall
right wall
top deck

M1: the shunt is internal between 2 (DC IN-) and 1 (LOAD-). Input negative goes to 2, never to W1 - on the bus it bypasses the shunt and the meter reads 0 A.

SW1: A from M1.4, B to the outputs, yellow to ground. Never swap B and yellow: dead short when switched on. Beep the pins before crimping.

W1 ports: 5 = LOAD-, 4 = RISER -, 3 = CARD -, 2 = OUT - post, 1 = PROBE GND. LED - rides on the OUT - post.

Supply: OWON at 12.0 V, OVP 12.6 V, 6 A current limit (the working max). F1 8 A fast 5x20. J4 is optional.

wiring/pcie_rig_power - rev C 2026-09-28, edit in KiCad
lamMrCupp / pcie-gpu-test-rig

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File: pcie_rig_power.kicad_sch

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